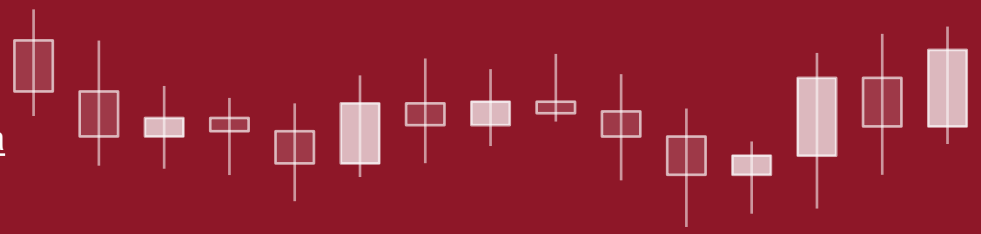
AI, MACHINE LEARNING AND BIG DATA FOR FINANCE AND MANAGEMENT · FINAL REPORT

Can Machine Learning Trade the Next Nasdaq Candle?

Next-candle direction models and long/short backtests on E-mini Nasdaq-100 futures (NQ), after commissions

Alessandro Pacifici
Waseda University · November 2026
Option 1: data analysis · Code and results: github.com/Paciooo/Project_ai_ML_bigdata



Executive Summary

Models predict slightly better than chance, but no strategy survives costs and robustness tests

6.2M

one-minute NQ bars (2006–2025), tested on 1, 5, 15, 60-minute and daily candles

+0.62 pp

best accuracy gain over “always up” (MLP 128-64-32, 5 min, 95% CI 0.32 to 0.92)

407

model × direction × filter strategies compared on validation data

0

strategies passing all tests: profitable out of sample, after costs, and robust

Problem

Can a model predict whether the next Nasdaq futures candle closes above its open, and trade that signal profitably after $5 round-trip commissions?

Approach

Original vs optimised logistic regression, XGBoost, random forest and neural networks (up to 4 hidden layers), new causal features, strict chronological splits, long/short, long-only and short-only backtests.

Verdict

Optimisation adds less than one point of accuracy. Intraday strategies lose after costs; daily profits mostly track the Nasdaq's upward drift. No deployable strategy was found.

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Task, Data and Models

Predict the body of the next candle at the close of the current one, then trade exactly that candle

Prediction task (no look-ahead)



- **Target:** bullish if close(t+1) > open(t+1), bearish if lower; doji = miss
- **Data:** NQ futures (CME E-mini Nasdaq-100, $20 per point), TradeStation 1-min bars 2006–2025
- **Candles:** 1, 5, 15, 60 min and daily (one CME session, 17:00–16:00 CT)
- **Splits:** train 2006–Aug 2017 · validation → Jul 2021 · test Jul 2021–May 2025, purged at boundaries

Model families: original and optimised versions

1. **Logistic regression**
   weighted sum of features → probability of an up candle; simple, hard to overfit
2. **Random forest**
   averages hundreds of decision trees built on random data subsets
3. **XGBoost**
   trees added one by one, each correcting the previous errors; early-stopped on validation
4. **Neural network (MLP)**
   stacked layers of neurons learning non-linear patterns; 2 to 4 hidden layers tested

Features: 8 original price features (lagged returns, momentum, volatility) + 11 new ones (candle body and wicks, range, volume, realised volatility, RSI, moving-average gaps, time of day).

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Original vs Optimised Models

Optimisation changes out-of-sample accuracy by less than one point; deeper networks give no consistent gain

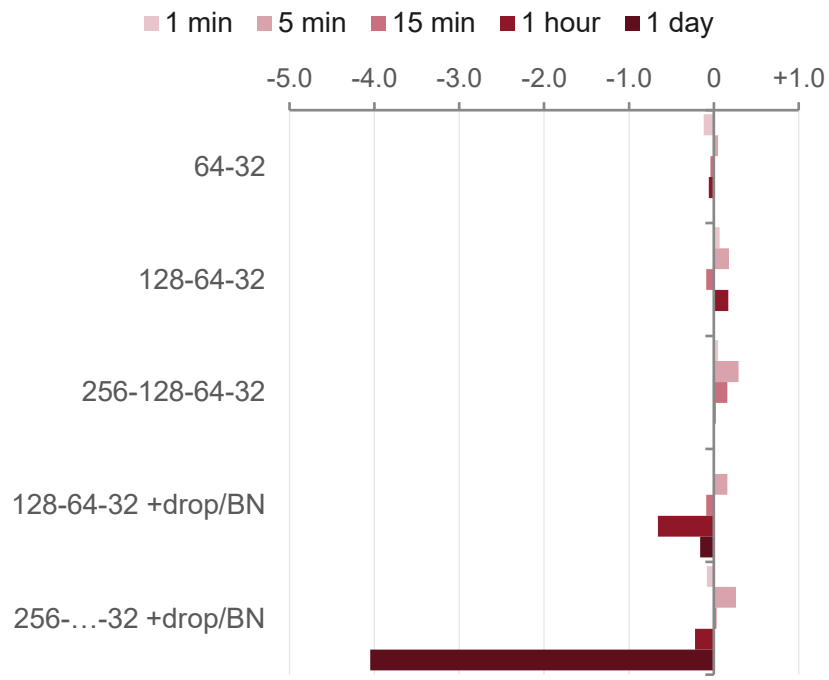
Test accuracy, all candles (original → optimised, change in pp)

Model	1 min	5 min	15 min	1 hour	1 day
Logistic regression	48.63% → 48.67% +0.04 pp	50.42% → 50.43% +0.01 pp	50.73% → 50.77% +0.05 pp	51.09% → 50.50% −0.59 pp	51.71% → 51.71% +0.00 pp
Random forest	48.68% → 48.69% +0.01 pp	50.49% → 50.47% −0.02 pp	50.88% → 50.79% −0.09 pp	51.11% → 51.69% +0.59 pp	54.21% → 54.21% +0.00 pp
XGBoost	48.63% → 48.63% −0.00 pp	50.50% → 50.46% −0.04 pp	50.67% → 50.96% +0.29 pp	50.77% → 51.76% +0.98 pp	51.56% → 51.56% +0.00 pp
Neural network (MLP)	48.66% → 48.71% +0.05 pp	**50.33% → 50.52% +0.18 pp**	50.94% → 50.85% −0.09 pp	51.57% → 51.74% +0.17 pp	54.21% → 54.21% +0.00 pp

Highlighted: best predictive model, MLP 128-64-32 on 5 min candles: 51.45% correct on non-doji candles, +0.62 pp vs always-up (95% CI 0.32 to 0.92). Dojis count as misses, so all-candle accuracy at 1 min is below 50%.

Accuracy beats "always up" only at 1 and 5 minutes; from 15 minutes on, no model is reliably better

Neural networks: test accuracy vs original 64-32-16, pp



Early stopping halted every network after 5–17 epochs: validation loss stops improving almost immediately, a sign of a very weak signal. More neurons or layers gave no consistent gain; dropout/batch norm did not help.

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From Signals to Trades

Unfiltered, every candidate loses after $5 per round trip: intraday the gross edge is below $5 a trade; daily there is no edge

Trading rules (one NQ contract)

1 — Signal
model probability that candle t+1 is bullish

2 — Direction
Long & Short · Long only · Short only

3 — Filter (optional)
trade only if |P(up) − 0.5| ≥ threshold chosen on validation

4 — Execution
enter at open(t+1), exit at close(t+1); P&L = ±(close − open) × $20

5 — Costs
$5 per round trip ($2.50 open + $2.50 close), zero slippage, 100% fills (idealised)

Validation-chosen model per candle length, Long & Short, test period

Candle	Model	Trades	Gross / trade	Net	Filtered trades	Filtered gross / trade	Filtered net
1 min	MLP 256-128-64-32	1,370,692	−$0.17	**−$7.09M**	2,163	+$19.19	**+$30.7k**
5 min	MLP 128-64-32, lr 3e-4	273,340	+$0.42	**−$1.25M**	9,717	−$0.90	**−$57.4k**
15 min	MLP 128-64-32, lr 3e-4	90,189	−$0.53	**−$498.8k**	1,082	+$2.74	**−$2.4k**
1 hour	XGBoost d5, early stop	21,507	+$1.52	**−$74.9k**	17,419	+$4.56	**−$7.6k**
1 day	MLP 256-128-64-32 + dropout/BN	642	−$60.61	**−$42.1k**	575	−$40.09	**−$25.9k**

Gross / trade = average P&L per trade before the $5 commission. The filter (threshold chosen on validation) cuts trades by 10% to 99.8%; what remains is close to zero and unreliable.

Example, 5-minute reversal rule: +$391.3k gross over 273,340 trades becomes −$975.4k after commissions: $1.43 gross edge per trade vs $5.00 cost

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Backtest by Model: Daily Candles

After costs 10 of 13 model × direction combinations are positive, mostly by riding the Nasdaq's rise

Test period, no filter, one contract, $5 per round trip (1 day candles chosen on validation)

Model	Accuracy	Net: Long & Short	Net: Long only	Net: Short only	Trades	Win rate	Gross / trade	Net / trade	Max drawdown
Logistic (original)	51.71%	+$110.0k	+$92.4k	+$17.6k	642	51.7%	+$176.31	+$171.31	−$70.3k
XGBoost (original)	51.56%	+$56.9k	+$66.1k	−$9.1k	642	51.6%	+$93.66	+$88.66	−$61.3k
Random forest d10	54.21%	+$73.0k	+$73.0k	no trades	642	54.2%	+$118.75	+$113.75	−$79.9k
MLP 64-32-16 (original)	54.21%	+$73.0k	+$73.0k	no trades	642	54.2%	+$118.75	+$113.75	−$79.9k
MLP 256-128-64-32 + dropout/BN	50.16%	−$42.1k	+$16.1k	−$58.2k	642	50.2%	−$60.61	−$65.61	−$100.9k
Always long	54.21%	—	+$73.0k	—	642	54.2%	+$118.75	+$113.75	−$79.9k

Trades, win rate, per-trade P&L and drawdown: Long & Short version (Long only for “always long”). Gross / trade is before the $5 commission. Highlighted: model of the pre-specified best strategy.

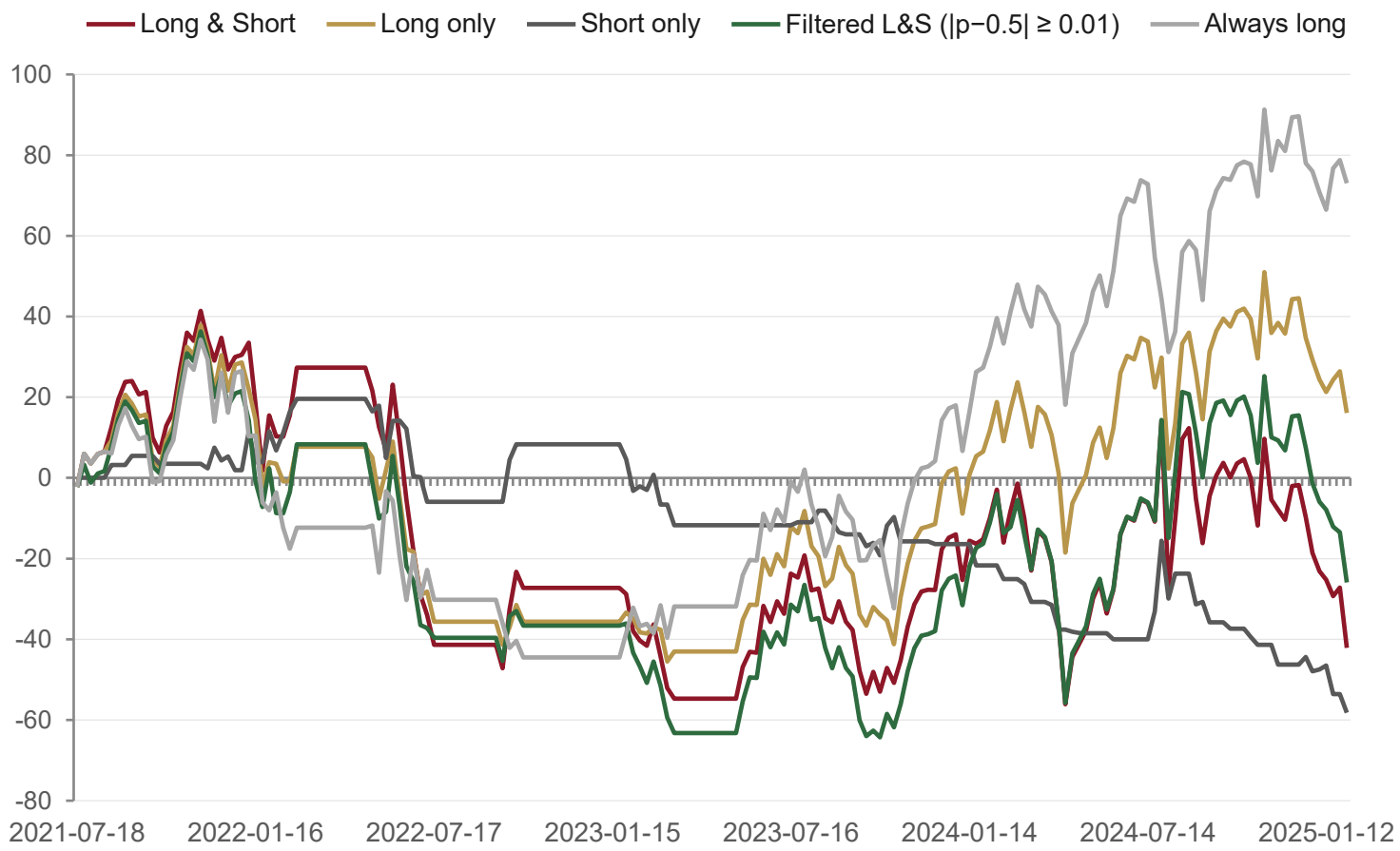
Staying long on every candle earned +$73.0k with no model at all; 1 of 5 models beat it with Long & Short, and none has a daily-P&L 95% interval above zero

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Equity Curves: Long vs Short

Long-only beats short-only, but no direction beats simply staying long (+$73.0k)

Cumulative net P&L, test period, $k: MLP 256-128-64-32 + dropout/BN, 1 day candles



Long vs short contribution (no filter)

	Long only	Short only
Trades	520	122
Net P&L	+$16.1k	−$58.2k
Win rate	52.7%	39.3%
Gross / trade	+$35.89	−$471.93
Max drawdown	−$83.4k	−$84.1k

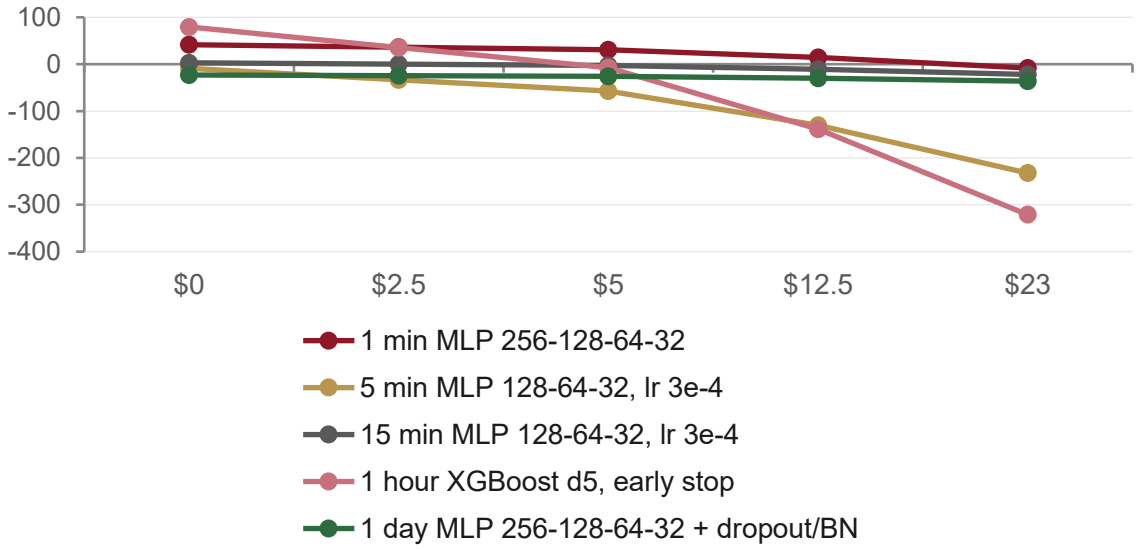
- The model calls “up” 81% of the time, so Long & Short is mostly a long position
- Short trades destroy value: −$58.2k net
- Long & Short: gross −$38.9k, commissions −$3.2k, net −$42.1k

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Robustness: Costs, Walk-Forward, In vs Out of Sample

Every candidate profitable on validation loses or shrinks on the test period

Test net P&L vs cost per round trip, filtered candidates (thousand USD)



$12.50 and $23 add one tick of spread plus ¼ or 1 tick of slippage per side: realistic market-order execution.

Net P&L by sample and walk-forward years (filtered, $5)

Candle	Train	Validation	Test	Walk-forward years > 0	Walk-forward net
1 min	+$1.7k	+$48.8k	**+$30.7k**	2 of 4	+$3.7k
5 min	−$63.1k	+$81.3k	**−$57.4k**	1 of 4	−$16.7k
15 min	+$6.8k	+$51.0k	**−$2.4k**	1 of 4	−$65.5k
1 hour	+$63.2k	+$133.8k	**−$7.6k**	2 of 4	−$157.5k
1 day	+$102.6k	+$164.5k	**−$25.9k**	4 of 4	+$201.5k

Validation picks the best of 407 configurations, so its profit is optimistic by construction (multiple comparisons). Walk-forward: model refitted each year on all prior data, threshold re-chosen on the preceding year.

No candidate has a 95% interval of daily net P&L above zero in the test period. Only 1 day is profitable in ≥ 3 walk-forward years, with yearly-refitted models that are mostly long in a rising market; the fixed model loses on the test

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Best Overall Strategy

No deployable NASDAQ trading strategy has been identified

Selected by the pre-specified rule
highest validation profit across all 407 configurations

Model	MLP 256-128-64-32 + dropout/BN
Candles · direction	1 day · Long & Short
Confidence filter	|P(up) − 0.5| ≥ 0.01
Test accuracy	50.16%
Net P&L (test)	**−$25.9k**
Trades · win rate	575 · 50.6%
Avg trade: gross · net	−$40.09 · −$45.09
Profit factor*	0.97
Max drawdown	−$102.4k
Daily net 95% CI	−$351 to $267
Walk-forward	4 of 4 years positive, +$201.5k

Best test-period result (descriptive only)
chosen after seeing the test: not a valid selection

Model	MLP 256-128-64-32
Candles · direction	1 min · Long & Short
Confidence filter	|P(up) − 0.5| ≥ 0.1
Test accuracy	48.71%
Net P&L (test)	**+$30.7k**
Trades · win rate	2,163 · 52.1%
Avg trade: gross · net	+$19.19 · +$14.19
Profit factor*	1.10
Max drawdown	−$16.4k
Daily net 95% CI	−$11 to $85
Walk-forward	2 of 4 years positive, +$3.7k

* Profit factor = sum of winning trades ÷ |sum of losing trades|, both after commissions; 1.00 = winners exactly pay for losers. Avg trade gross = before the $5 commission.

Verdict: neither passes the pre-set bar (test profit at $5 and $12.50, daily-P&L interval above zero, 3 of 4 profitable walk-forward years). Trading either would be a bet, not an edge. The 1 day walk-forward profit reflects being mostly long in a rising market.

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Appendix

A1 Data, splits and sample sizes

A2 Features, hyper-parameters and selection rules

A3 Classification metrics and confusion matrix

A4 Net P&L by candle length, model and direction

A5 Why 50% means random

A6 Reproducibility and limitations

A1 Data, Splits and Sample Sizes

Same chronological boundaries for every candle length; test scored once

Candle	Rows (all)	Train	Rows fitted	Validation	Test	Test dojis	Test bullish share
1 min	5,884,299	3,167,361	2,000,000	1,346,246	1,370,692	58,616	50.4%
5 min	1,312,758	770,236	653,403	269,182	273,340	4,933	50.8%
15 min	442,614	264,084	240,005	88,341	90,189	909	51.3%
1 hour	108,026	64,877	62,096	21,642	21,507	104	51.8%
1 day	3,457	1,991	1,983	822	642	0	54.2%

- Train 2006-01-03 → 2017-08-24, validation → 2021-07-14, test 2021-07-15 → 2025-05-30; labels crossing a boundary are purged. Test seen in earlier phases: exploratory.
- Data: TradeStation continuous NQ 1-minute bars, additively back-adjusted (point moves equal the traded contract's).
- NQ = $20 × Nasdaq-100 index, tick 0.25 = $5; notional ≈ $210k–$445k per contract in 2021–2025. P&L = points moved × $20.
- Candles stay inside one CME session (daily = 17:00–16:00 CT). Bad prints (|1-min return| > 2%) quarantine the next 50 candles.
- Daily: early-close sessions are not targets; the quarantine after 3 Feb 2025 ends the daily test on 2025-01-30.

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A2 Features, Hyper-parameters and Selection Rules

All choices made on training and validation data; the test set never informed them

Features (all known at the close of candle t)

Group	Features
Original (8)	log returns lag 1, 2, 3, 5; momentum 5, 10; volatility 5, 10
Candle shape	body / close, high-low range, upper and lower wick
Volume, volatility	log volume vs 20-bar mean; 20-bar realised volatility
Indicators	RSI(14) centred; gap to 20 and 50-bar moving averages
Time	time of day within the session (sine, cosine)

Configurations tested per candle length

Family	Original	Variants (new features)
Logistic	C = 1, 8 features	C = 1 and C = 0.01
Random forest	200 trees, depth 6	200 trees, depth 10, leaf 200
XGBoost	200 trees, depth 3, η 0.05	depth 3 or 5, η 0.05, early stop on validation
MLP	64-32-16, ReLU, L2 1e-3	64-32, 128-64-32, 256-128-64-32; ±dropout 0.2 + batch norm; lr 3e-4 / batch 1024; AdamW, early stop

- Optimised model per family = highest validation balanced accuracy (it may be the original).
- Confidence threshold per model and direction = highest validation net P&L on {0, 0.01, 0.02, 0.03, 0.05, 0.075, 0.10}, at least 200 trades.
- Overall strategy = highest validation net P&L; deployable only if test net > 0 at $5 and $12.50, daily-P&L 95% interval > 0 and ≥ 3 of 4 profitable walk-forward years.
- Scaling fitted on training rows only; network training shuffles only inside the training set; seeds fixed (42). Daily networks use batches of 128 (≈2,000 training rows).
- Profit factor = sum of winning trades ÷ |sum of losing trades|, after commissions. Gross / trade = average P&L per trade before the $5 commission.

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A3 Classification Metrics and Confusion Matrix

Test period, 5 min candles: originals and optimised versions

Model	Accuracy	Non-doji	Balanced	Precision (up)	Recall (up)	F1 (up)	Bullish acc.	Bearish acc.	vs majority, pp [95% CI]
Always long	49.90%	50.81%	50.00%	50.81%	100.00%	67.39%	100.00%	0.00%	0.00 [0.00, 0.00]
Reversal rule	50.30%	51.23%	51.22%	52.04%	51.27%	51.65%	51.27%	51.18%	0.40 [0.12, 0.68]
Logistic (original)	50.42%	51.35%	51.16%	51.76%	62.85%	56.76%	62.85%	39.48%	0.53 [0.25, 0.80]
Logistic C=1 + features	50.43%	51.36%	51.28%	51.99%	55.90%	53.87%	55.90%	46.67%	0.53 [0.27, 0.79]
XGBoost (original)	50.50%	51.43%	51.33%	52.00%	57.43%	54.58%	57.43%	45.23%	0.60 [0.32, 0.91]
XGBoost d3, early stop	50.46%	51.39%	51.32%	52.03%	55.73%	53.81%	55.73%	46.91%	0.57 [0.27, 0.86]
Random forest (original)	50.49%	51.42%	51.32%	52.00%	57.33%	54.53%	57.33%	45.32%	0.60 [0.30, 0.88]
Random forest d10	50.47%	51.40%	51.34%	52.07%	54.93%	53.46%	54.93%	47.75%	0.57 [0.28, 0.88]
MLP 64-32-16 (original)	50.33%	51.26%	51.12%	51.78%	59.48%	55.36%	59.48%	42.77%	0.44 [0.16, 0.71]
MLP 128-64-32	50.52%	51.45%	51.42%	52.18%	53.16%	52.67%	53.16%	49.68%	0.62 [0.32, 0.92]

Confusion matrix, MLP 128-64-32 (non-doji candles)

	Predicted up	Predicted down
Actual up	72,499	63,891
Actual down	66,431	65,586

n = 273,340 test candles (136,390 bullish, 132,017 bearish, 4,933 doji). Accuracy ± 0.19 pp (binomial 95%); the paired comparison with the majority rule uses 20-session block bootstrap.

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A4 Net P&L by Candle Length, Model and Direction

Test period, $5 per round trip, thousand USD: unfiltered (left) and with the validation-chosen filter (right)

Candle	Model (optimised)	L&S	Long	Short	L&S filtered	Long filtered	Short filtered
1 min	Logistic C=1 + features	-7715.1	-4023.4	-3691.7	-70.6	-45.5	-25.1
	Random forest d10	-7366.8	-4066.8	-3300.0	-3.3	-3.2	-65.2
	XGBoost d5, early stop	-7393.0	-4122.6	-3270.3	-14.0	-11.5	-2.5
	MLP 256-128-64-32	-7088.5	-3909.7	-3178.8	30.7	2.2	28.5
5 min	Logistic C=1 + features	-1158.5	-630.8	-527.7	36.1	-16.1	44.0
	Random forest d10	-1045.1	-560.1	-485.0	-5.5	10.1	-15.6
	XGBoost d3, early stop	-1135.4	-616.3	-519.1	29.9	14.6	15.3
	MLP 128-64-32	-1138.5	-581.6	-556.9	-26.1	-3.5	-9.8
15 min	Logistic C=0.01 + features	-557.1	-317.7	-239.4	-16.6	-36.6	20.0
	Random forest d10	-335.6	-200.6	-135.0	4.0	3.8	13.7
	XGBoost d3, early stop	-408.7	-251.7	-157.0	-15.6	-18.1	44.8
	MLP 128-64-32	-515.1	-290.3	-224.9	-52.5	-60.8	8.2
1 hour	Logistic C=1 + features	-128.9	-89.2	-39.8	-189.4	-201.3	-20.0
	Random forest d10	-5.2	-31.0	25.7	-61.4	—	-0.5
	XGBoost d5, early stop	-74.9	-69.1	-5.8	-7.6	-29.1	21.5
	MLP 128-64-32	-130.0	-92.5	-37.6	-137.6	-106.9	-30.7
1 day	Logistic (original)	110.0	92.4	17.6	92.3	68.2	—
	Random forest d10	73.0	73.0	0.0	—	—	—
	XGBoost (original)	56.9	66.1	-9.1	92.2	76.7	15.6
	MLP 64-32-16 (original)	73.0	73.0	0.0	—	—	—

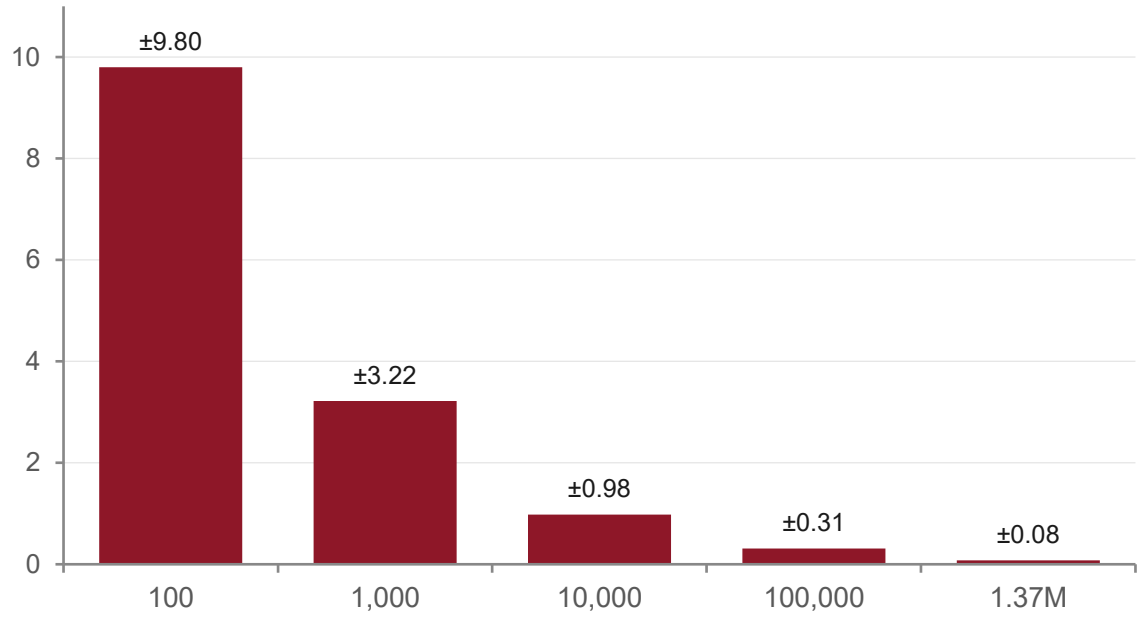
"—": the validation-chosen threshold was 0 (no filter) or produced fewer than 200 validation trades. Intraday filtered results are spread around zero; daily results are positive mainly because the models stay long.

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A5 Why 50% Means Random

Predicting the next candle is close to tossing a coin; only results outside luck's range count

How far luck alone moves a fair coin's accuracy (95% range, ± pp)



Range = 1.96 × √(0.25 / n). With few trades a coin can look like a 55–60% "model"; with a million it cannot leave 50 ± 0.1%.

Reading accuracy around 50%

1 **Coin toss = 50%**
if up and down are equally likely and unpredictable, any rule is right half the time

2 **Dojis lower the bar**
candles with close = open count as misses, so all-candle accuracy can sit below 50%

3 **Imbalance is not skill**
NQ rose over time, so "always up" already beats a coin without predicting anything

4 **Accuracy ≠ money**
a 51% hit rate earns nothing if wins and losses are equal in size and each trade costs $5

Best model: 51.45% on non-doji 5 min candles: statistically real, economically too small

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A6 Reproducibility and Limitations

Every number in this deck is regenerated and cross-checked by four commands from the public repository

Reproduce (Windows PowerShell, project folder)

Repository: github.com/Paciooo/Project_ai_ML_bigdata

```
$env:FUTURES_DATA_DIR="C:\path\to\futures-data"
```

```
1. python study.py
```

trains all models, backtests, walk-forward (about 15 min on 16 CPU cores, no GPU)

this deck: 1 min, 5 min, 15 min, 1 hour run + 1 day run, joined by tools/merge_runs.py

```
2. python report.py results/nq_study/runs/<run>
presentation/deck_data.json
```

```
3. python verify.py results/nq_study/runs/<run>
```

```
4. cd presentation; npm install; node build_deck.js
```

Runs used here: 20261111T070520Z_949daa_final (commit ff445f3) + 20261111T103357Z_02bd49_daily_final (commit 0970132); seed 42, latency 0s, $5 per round trip.

Limitations

- Idealised execution: fill at the next open with zero delay and zero slippage, 100% limit fills; a live limit order may not fill.
- Bar prices, not bid/ask quotes; costs are assumptions. $12.50 / $23 scenarios approximate market orders.
- Continuous back-adjusted series; roll dates are approximate (no position spans a session).
- The 2021–2025 test period was seen in earlier phases; data after May 2025 (reserved) was not available.
- Daily Sharpe ratios use 252 sessions; percentages use an assumed $100,000 account.

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